

JOINT PROBABILITY DISTRIBUTIONS

So far, we studied probability distributions for a single random variable. However, in many random experiments there are more than one random variable involved. For example, the length and width of a randomly selected rectangular cover, the position (X, Y) of a particle moving at random on the XY -plane, the speed and size of the wind in a hurricane, and so on.

In general, if X, Y are two random variables, the probability distribution that defines their simultaneous behavior is called a **joint probability distribution**.

I. Joint Probability Mass Function

Let X, Y be two discrete random variables. In this case, the joint probability distribution is a description of the set of points (x, y) along with their probabilities.

Notation. The probability that $X = x$ and $Y = y$ is written as

$$P(X = x, Y = y)$$

The **joint probability mass function** of discrete random variables X, Y denoted $f(x, y)$ is a function that satisfies the following conditions:

- (i) $f(x, y) \geq 0$ for all (x, y)
- (ii) $\sum_X \sum_Y f(x, y) = 1$

In this case,

$$P(X = x, Y = y) = f(x, y)$$

As one can see, these conditions are predictable generalizations of the conditions given in the one variable case. The first condition implies that the function is nonnegative for all possible values of (x, y) , and the second one implies that the sum of all values is equal to one. Of course, like in one variable case, the values of function is assumed to be zero at all values outside of its range.

EXAMPLE

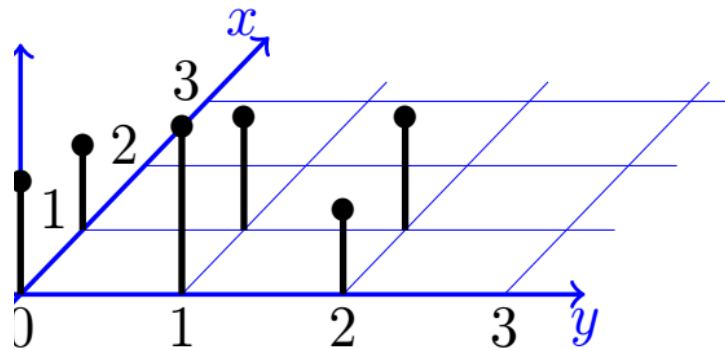
Suppose a function $f(x, y)$ is defined by the following table:

$Y \backslash X$	0	1
0	$\frac{1}{6}$	$\frac{1}{12}$
1	$\frac{1}{4}$	$\frac{1}{6}$
2	$\frac{1}{12}$	$\frac{1}{4}$

Then this is indeed a joint probability mass function since all values of $f(x, y)$ are positive and they add up to one:

$$\frac{1}{6} + \frac{1}{4} + \frac{1}{12} + \frac{1}{12} + \frac{1}{6} + \frac{1}{4} = \frac{12}{12} = 1$$

Below, is the graph of this function. Notice that, the way we graph a joint probability mass function is also similar to the one variable case.



So in this example,

$$P(X = 0, Y = 0) = f(0, 0) = \frac{1}{6}$$

$$P(X = 1, Y = 2) = f(1, 2) = \frac{1}{4}$$

II. Joint Probability Density Function

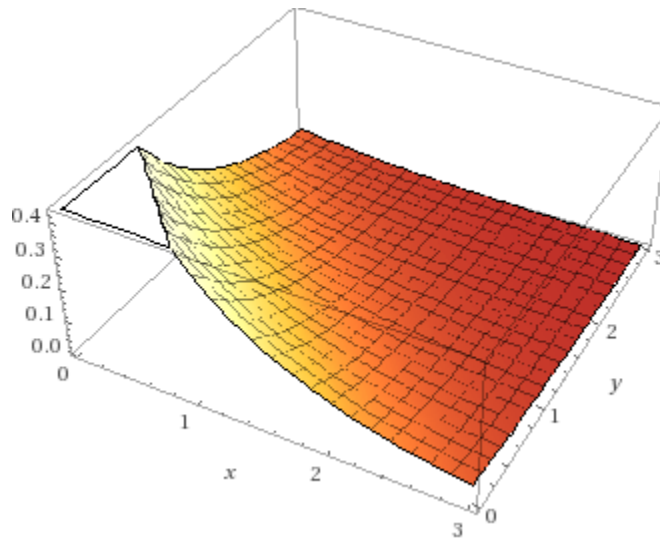
The joint probability density function gives us the probability of two continuous random variables X, Y to be in a region D in the two-dimensional space.

The definition is very similar to that of one variable case:

Let X, Y be two continuous random variables. Then a function $f(x, y)$ is said to be a **joint probability density function** if it satisfies the following conditions:

- (i) $f(x, y) \geq 0$ for all x, y
- (ii) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$

So, this function is always nonnegative and the total volume under the surface defined by this function is one. The values of function is assumed to be zero at all values outside of its range. Below is the graph of a joint probability density function:



Note that for any region D in the XY –plane

$$P((X, Y) \in D) = \iint_D f(x, y) dx dy$$

Marginal Probability Density Function

In both the discrete and the continuous cases, we may be interested in the probability distribution of each variable individually. This individual probability distribution is called the ***marginal probability distribution***.

In the discrete case, to determine $P(X = x)$, we just find the sum over all values of Y corresponding to $X = x$. For example, let the joint probability mass function be defined as

$Y \backslash X$	0	1
0	$1/6$	$1/12$
1	$1/4$	$1/6$
2	$1/12$	$1/4$

Then the marginal probability

$$P(X = 1) = \frac{1}{12} + \frac{1}{6} + \frac{1}{4} = \frac{1}{2}$$

Continuous Case

Let X, Y be two continuous random variables with joint probability density function $f(x, y)$. Then, ***the marginal probability density function of X*** is denoted as $f_X(x)$ and is defined as

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

Similarly, ***the marginal probability density function of Y*** is denoted as $f_Y(y)$ and is defined as

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Note that f_X is only a function of x and f_Y is only a function of y .

Common Joint Distributions

The Multinomial Distribution

The binomial distribution can be generalized to generate a useful joint probability distribution for multiple discrete variables. The random experiment consists of a series of independent trials where each trial can be categorized into one of k classes. The resulting distribution is called the ***multinomial distribution***.

Suppose a random experiment consists of a series of n trials such that

- (i) The result of each trial is classified into one of k classes.
- (ii) The probability of a trial generating a result in class 1, class 2, ..., class k is given as $p_1, p_2, \dots, p_k$, respectively, and these probabilities are constant over the trials.
- (iii) The trials are independent.

The random variables $X_1, X_2, \dots, X_k$ that denote the number of trials that result in class 1, class 2, ..., class k , respectively, is said to have a ***multinomial distribution*** and the joint probability mass function is given as

$$P(X_1 = x_1, X_2 = x_2, \dots, X_k = x_k) = \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

Example

Bits are transmitted independently. When bits are received, there is 0.6 probability that they will be excellent, 0.3 that they will be good, 0.08 that they will be fair, and 0.02 that they will be poor. Of the 20 -bits received, what is the probability 14 are excellent, 3 are good, 2 are fair, and 1 is poor?

$$P(X_1 = 14, X_2 = 3, X_3 = 2, X_4 = 1) = \frac{20!}{14! \times 3! \times 2! \times 1!} (0.6)^{14} (0.3)^3 (0.08)^2 (0.02)^1 =$$

Bivariate Normal Distribution

Recall that in one variable case the probability density function is given as

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

The probability density function of a **bivariate normal distribution** is

$$f(x, y; \sigma_X, \sigma_Y, \mu_X, \mu_Y, \rho)$$

$$= \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \times \exp\left\{\frac{-1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2}\right]\right\}$$

for $-\infty < x < \infty, -\infty < y < \infty$. The parameters satisfy the conditions $\sigma_X > 0, \sigma_Y > 0, -\infty < \mu_X < \infty, -\infty < \mu_Y < \infty$, and $-1 < \rho < 1$. The quantity ρ is related to the correlation between X and Y . In fact,

$$\rho = \frac{\text{cov}(X, Y)}{\sigma_X\sigma_Y}$$

It can be shown that this function integrates to 1 and is positive over the entire plane.

The bivariate normal probability density function is constant on ellipses in the (x, y) plane. The center of each ellipse is at the point (μ_X, μ_Y) . If $\rho > 0$, the major axis of each ellipse has a positive slope and if $\rho < 0$, they have negative slopes. If $\rho = 0$, the major axis is aligned with either the x - or the y -axis.

Its marginal distribution with respect to X is $N(\mu_X, \sigma_X)$ and its marginal distribution with respect to Y is $N(\mu_Y, \sigma_Y)$.

To solve probability problems, we must integrate the joint pdf over appropriate regions. Unfortunately, there is no closed-form solution to such problems and integrations must be done numerically.

Here is a simple example.

Example. Let, X, Y be jointly normal random variables with $E(X) = \mu_X = 0$, $E(Y) = \mu_Y = -1$, $Var(X) = \sigma_X^2 = 1$, $Var(Y) = \sigma_Y^2 = 4$, and $\rho = -\frac{1}{2}$. Find $P(X + Y > 0)$.

If X, Y normal, then $U = X + Y$ is normal. Also,

$$E(U) = E(X + Y) = E(X) + E(Y) = 0 + (-1) = -1$$

Similarly,

$$Var(U) = Var(X + Y) = Var(X) + Var(Y) + 2cov(X, Y)$$

We have

$$cov(X, Y) = \rho\sigma_X\sigma_Y = \left(-\frac{1}{2}\right)(1)(2) = -1$$

Thus,

$$Var(U) = Var(X + Y) = Var(X) + Var(Y) + 2cov(X, Y) = 1 + 4 + 2(-1) = 3$$

Hence, $U = X + Y$ is a normal distribution with expected value -1 and standard deviation $\sqrt{3}$. So, using our calculator,

$$P(X + Y > 0) = P(U > 0) = \text{normcdf}(-100, 0, -1, \sqrt{3}) = 0.2818$$

Bivariate Distributions in R

Loading the Packages

We need to load the packages *bivariate* and *MASS*.

```
>install.package ("bivariate")
```

```
>install.package ("MASS")
```

```
> library (bivariate)
```

```
> library (MASS)
```

Note that the *bivariate* package imports the *barsurf*, *mvtnorm* and *KernSmooth* packages.

Discrete Bivariate Uniform Distributions

We can describe a bivariate uniform distribution as the product of two univariate uniform distributions. To construct the probability mass function, we use *dubvpmf* (). The arguments are the a_X , the lower limit for X ; b_X , the upper limit for X ; a_Y , the lower limit for Y ; and b_Y , the upper limit for Y .

```
>f = dubvpmf( $a_X, b_X, a_Y, b_Y$ )
```

To graph this function we write

```
>plot(f, TRUE)
```

To construct the cumulative distribution function, we can use the *dubvcdf*() function. It takes the same arguments as *dubvpmf*().

```
> F = dubvcdf ( $a_X, b_X, a_Y, b_Y$ )
```

and graphing is done the same way:

```
> plot (F, TRUE)
```

Bivariate Binomial Distributions

Suppose we have n trials. In each trial, there are two independent events, each with a fixed probability of success. Assume moreover, the events are independent. An example would be flipping two coins, n times.

Suppose we have random variables X and Y counting the number of successes. Let p_X and p_Y be the successes, respectively. We can construct its probability mass function using the ***bnbvpmf()*** function. It takes three arguments, the probability of the first success, the probability of the second success and the number of trials.

```
> f = bnbvpmf (p_X, p_Y, n)
```

We can also graph the distribution

```
> plot (f, TRUE)
```

To construct the cumulative distribution function, we can use the ***bnbvCDF()*** function. It takes the same arguments as ***bnbvpmf()***.

```
> F = bnbvCDF (p_X, p_Y, n)
```

Graphing works the same way:

```
> plot (F, TRUE)
```

Bivariate Poisson Distributions

Based on Karlis and Ntzoufras (2003), we can define the bivariate Poisson probability mass function as:

$$P(X = x, Y = y) = f(x, y, \lambda_1, \lambda_2, \rho) = e^{-(\lambda_1 + \lambda_2 + \rho)} \frac{\lambda_1^x \lambda_2^y}{x! y!} \sum_{k=0}^{\min(x, y)} \binom{x}{k} \binom{y}{k} k! \left(\frac{\rho}{\lambda_1 \lambda_2} \right)^k$$

where

$$E(X) = \text{var}(X) = \lambda_1 + \rho$$

$$E(Y) = \text{var}(Y) = \lambda_2 + \rho$$

and ρ is the correlation between X and Y .

For the probability mass function, we can use the ***pbvpmf()*** or ***pbvpmf.2()*** functions. The first takes three arguments: $\lambda_1, \lambda_2, \rho$. The second also takes three arguments, but they are $E(X), E(Y)$ and ρ .

So we can write

```
> f = pbvpmf ( $\lambda_1, \lambda_2, \rho$ )
```

or

```
> f = pbvpmf.2( $E(X), E(Y), \rho$ )
```

Graphing works the same way:

```
> plot (f, TRUE)
```

To construct the cumulative distribution function, we can use the ***pbvcdf()*** or ***pbvcdf.2()*** functions.

They take the same arguments as ***pbvpmf()*** and ***pbvpmf.2()***.

```
> F = pbvcdf ( $\lambda_1, \lambda_2, \rho$ )
```

or

```
> F = pbvcdf.2 ( $E(X), E(Y), \rho$ )
```

and to graph we write

```
> plot (F, TRUE)
```

Note that λ_1, λ_2 need to be greater than zero.

Continuous Bivariate Uniform Distributions

Continuous bivariate uniform distributions are similar to discrete bivariate uniform distributions. However, we have a probability density function instead of a probability mass function.

We can use the ***cubvpdf()*** function. The arguments are the a_X , the lower limit for X ; b_X , the upper limit for X ; a_Y , the lower limit for Y ; and b_Y , the upper limit for Y .

```
> f = cubvpdf (a_X, b_X, a_Y, b_Y)
```

```
> plot (f, TRUE)
```

To construct a cumulative distribution function, we can use the ***cubvcdf()*** function. It takes the same arguments as ***cubvpdf()***.

```
> plot (F, TRUE)
```

Bivariate Normal Distributions

We can construct a normal bivariate probability density function using the ***nbvpdf()*** or ***nbvpdf.2()*** functions, both of which use the ***dmvnorm()*** function from the ***mvtnorm*** package, and take five arguments.

The first takes the mean of X , the mean of Y , the standard deviation of X , the standard deviation of Y , and their correlation. The second takes the mean of X , the mean of Y , the variance of X , the variance of Y and their covariance.

```
> f = nbvpdf (0, 0, 1, 1, 0)
```

And plot it.

```
> plot (f, TRUE)
```

We can construct a cumulative distribution function using the ***nbvcdf()*** or ***nbvcdf.2()*** functions, both of which use the ***pmvnorm()*** function from the *mvtnorm* package, and take the same arguments as ***nbvpdf()*** and ***nbvpdf.2()***.

Once we have constructed our object we can plot it.

```
> plot (F, TRUE)
```